

Macro News and a Futures Mapping Pipeline

A review of capabilities, reasoning, and limitations

April 14, 2026

Experiment background

Example mappings

Company-specific gate

Aggregate diagnostics

Cost & scale

Experiment background

Experiment background

Dataset

- Dow Jones Newswires, March 2022 (shards a–e)
- 1,000 articles, random sample with fixed seed
- Month chosen for dense macro signal: Russia–Ukraine, Fed liftoff, commodity shock

Asset universe

- 95 futures contracts across 7 asset classes
- Commodities, FX, government bonds, STIR, equity indices, US equity sectors, volatility

Model & infra

- Gemma 4 26B A4B Instruct via vLLM 0.19
- Deterministic ($T=0$), `max_model_len=65,536`
- 1× RTX Pro 6000 Blackwell, ~25 min end-to-end

Four-stage LLM pipeline

1. **Summarize** — macro summary + company-specific gate
2. **ArticleMapper** — whole article → (asset, signal, score)
3. **ParagraphMapper** — each paragraph, with macro summary as context
4. **Validator** — reviews union, accepts/rejects each pair

Run totals

- 726 / 1000 filtered as company-specific (73%)
- 4,157 (article, asset) candidates evaluated
- 4,019 accepted / 138 rejected (3.3% reject)
- Signal mix: 727 strong / 3,292 weak

Example mappings

Example 1 — BOE rate decision

Headline: *BOE's Rate Decision Could Benefit Sterling in Short-Term — Market Talk*

(DJNW, 2022-03-16, shard c)

Full article text (single “Market Talk” blurb)

1119 GMT — Sentiment around sterling looks quite downbeat but the Bank of England's meeting on Thursday may offer the currency some brief relief, HSBC says. A 25 basis-point rate rise is fully priced so that's unlikely to have a notable impact on sterling but the currency will rise in the short-term if the split of votes on the decision shows some policymakers backing a larger increase, HSBC forex analyst Dominic Bunning says in a note. “Should we see further calls for 50bp hikes this week by some Monetary Policy Committee members, then we would expect GBP to bounce.” The BOE raised its key rate by 25 basis points to 0.50% in February but four out of nine officials voted for a 50 basis-point rise. (renae.dyer@wsj.com)

Sterling

[currency]

strong, 0.95, both

Transmission: hawkish MPC split → higher expected terminal rate → wider rate differential → GBP up.

3-Month SONIA

[STIR]

strong, 0.95, both

Transmission: 50bp-vote signal directly repriced into the SONIA curve; SONIA is the sterling overnight benchmark.

UK Long Gilt

[government bond]

strong, 0.75, both

Transmission: more hawkish BoE path lifts gilt yields on the long end via the expectations and term-premium components.

Why this is a clean showcase: one trigger, three asset classes, both mappers independently agree, validator accepts all three.

The pipeline correctly scores the gilt lower — a real but less direct channel than FX/STIR.

Example 2a — USDA soybean export data (article text)

Headline: *U.S. Export Sales and Shipments: Soybeans — Mar 17*

(DJNW, 2022-03-17, shard c)

[para 0, caption] *For the week ended Mar 10 in thousand metric tons. Source: USDA. The current marketing year for soybeans began Sep 1.*

[para 1, data table] *Totals may not add due to rounding.*

Country	2021–2022 (current MY)				2022–23	
	Wk Δ com.	Wk ship.	Seas. ship.	Seas. com.	Wk Δ com.	Seas. com.
CHINA	395.5	370.2	25,626.3	27,686.3	406.0	5,639.0
EGYPT	241.8	61.8	2,316.7	3,349.7	0	105.0
TAIWAN	162.5	22.2	915.4	1,212.6	0	0
INDONESIA	83.2	87.7	922.3	1,201.3	0	0
MEXICO	56.2	120.6	2,812.9	4,805.6	0	123.5
... 40+ other countries, most 0						
TOTAL KNOWN	985.6	714.3	42,325.5	49,456.2	406.0	6,115.8
GRAND TOTAL	1,253.2	714.3	42,325.5	53,624.0	477.0	8,123.8

Fully structured data release — no prose “trigger” sentence. The pipeline must read the table itself.

Example 2b — reasoning & transmission channel

Soybeans [commodity] **strong, 0.95, both** **AM: 0.95** **PM(max): 0.95 (para 1)** **evidence: [1]**

ArticleMapper reasoning

USDA weekly export shipment and commitment data is a primary fundamental driver for CBOT Soybean futures. It shows “Week’s net Shipments” of **985.6k MT** and “Season’s Total Shipments” of **42,325.5k MT**. It highlights significant activity in major demand centers like **China (395.5k MT)** and **Egypt (241.8k MT)**. These data points directly impact the global supply/demand balance and the pace of US soybean exports.

ParagraphMapper [1] reasoning — 0.95 (paragraph 0 only got 0.50)

The trigger is the “total known shipments of 985.6k MT” and the high-volume net export activity in China (395.5k MT) and Egypt (241.8k MT). Transmission: high weekly shipments indicate strong international demand and reduce immediate domestic supply, exerting upward pressure on soybean futures prices.

Validator reasoning (accepted)

Direct report of USDA weekly data. The mappers correctly identified the China/Egypt figures and total known shipments as the primary triggers. *“I agree with the ArticleMapper’s assessment of a strong signal and high relevance.”*

Takeaways: (1) the pipeline reads a pure-data table and extracts the right country-level figures; (2) **evidence localization works** — caption para 0 = 0.50, data para 1 = 0.95; (3) all three stages converge independently.

Example 3a — Russia aerospace sanctions (article text)

Headline: *Heard on the Street: Russia Can't Fly Without The West* — WSJ

(DJNW, 2022-03-09, shard b)

Public link: [wsj.com/.../russia-cant-fly-without-the-west...](https://www.wsj.com/.../russia-cant-fly-without-the-west...)

Relevant article paragraphs

[3] Russia operates 3% of the global fleet, representing a steady stream of “aftermarket” cash for Western manufacturers. Boeing and Airbus have engineering centers there, which have now suspended operations. The U.S. company said Monday it had stopped buying from its Russian partner **VSMPO-AVISMA, the supplier of a quarter of the world's titanium** — a key input for airframes.

[4] Western lessors own **70% of Russia's 981 jets** in service or in storage and have no clear way of repossessing them. But the hit is deeply asymmetrical. **Only 1% of Airbus' and Boeing's order backlogs is affected.**

[7] Yet both use a lot of Western parts and engines: The MC-21 flies geared turbofans built by Pratt & Whitney. While work to “russify” the plane is already in the works, it would take decades. . . China, which has invested heavily in a similar plane, the C919, is the obvious partner.

A rich, narrative WSJ column. Multiple macro angles: a commodity supply shock (titanium), aerospace-manufacturer revenue loss, and a long-horizon Russia–China technology pivot. The pipeline tags 6 assets and rejects a seventh — we will walk through all of them.

Example 3b — all 9 mappings the pipeline produced

Asset	Class	Signal	Score	Source	Channel (pipeline's own words)
Materials Sector	US equity sector	strong	0.75	both	titanium supply shock (25% world supply)
Industrials Sector	US equity sector	weak	0.50	both	Boeing aerospace input costs
Euro Stoxx 50 Index	equity index	weak	0.45	article	Airbus heavyweight in Eurostoxx
Gold	commodity	weak	0.45	para	geopolitical risk premium
Information Tech. Sector	US equity sector	weak	0.40	article	Pratt & Whitney / MC-21 supply chain
CAC 40 Index	equity index	weak	0.40	article	Airbus (CAC 40 constituent)
Mini Nikkei 225 Index	equity index	weak	0.35	article	Japanese aerospace component exposure
DAX Index	equity index	weak	0.35	article	German aerospace supply chain
Hang Seng Index	equity index	weak	0.35	article	Russia–China bloc / C919 partnership

What to notice:

- **Only one strong tag.** Materials Sector gets promoted because titanium is a quantifiable, article-specific driver. Everything else stays weak.
- Weak mappings follow the actual **economic neighborhood**: Boeing's Industrials, Airbus-heavy Euro indices (CAC 40, DAX, Euro Stoxx 50), Japanese aerospace via Nikkei, HSI via the C919 pivot.
- **No runaway tag.** The article caps the impact at 1% of Boeing/Airbus order backlogs; the pipeline respects that — every weak mapping sits in a tight 0.35–0.50 score band.

Example 3c — the Materials Sector reasoning, verbatim

Materials Sector — strong, 0.75, both (accepted)

ArticleMapper: “The article identifies a specific supply chain disruption involving a critical industrial input: titanium. Boeing stopped buying from its Russian partner VSMPO-AVISMA, the supplier of a quarter of the world’s titanium — a key input for airframes. This creates a direct supply shock to the global titanium market. Because titanium is a fundamental raw material for the aerospace and materials sectors, a sudden loss of 25% of global supply is a significant trigger. . .”

ParagraphMapper [3]: “Boeing’s decision to stop purchasing from VSMPO-AVISMA, a supplier responsible for 25% of the world’s titanium. The transmission channel is a direct supply-side shock to a critical raw material used in airframe manufacturing. The Materials sector in US equities is heavily weighted toward producers and processors of industrial metals. . .”

Validator: “A direct supply-side shock to a critical commodity used in aerospace manufacturing. The transmission channel to the US Materials sector is sound.”

Why this example really sells the pipeline:

- **Non-trivial domain routing.** The article is about Boeing (Industrials), but the *strong* tag goes to **Materials** via titanium — the pipeline follows the *economic mechanism* not the company’s home sector, and independently across three stages (ArticleMapper, ParagraphMapper, Validator) converges on the same conclusion.
- **Quantified trigger drives the signal strength.** “25% of world titanium” is specific and large, so Materials gets 0.75 strong. The same article’s aerospace angle drives Industrials/CAC 40/DAX but only to weak 0.35–0.45, because those channels are indirect.
- **Calibrated breadth overall.** Nine tags, one strong, zero rejected — a clean consensus that this article is moderately relevant to a specific neighborhood of aerospace-exposed assets, with titanium as the one standout.

Example 4a — China property (article text)

Headline: *China Jan-Feb Property Investment Rose 3.7% on Year (+3 updates)*

(DJNW, 2022-03-14, shard b)

Full article text

[0] The downturn in China's property sector extended into early 2022 as Beijing's curbs on borrowing in the sector triggered uncertainty on the part of developers, while tight credit policies continued to pressure home buyer demand. Home sales by volume, a key indicator of demand, **plunged 22.1%** in the first two months of 2022 from a year earlier. This compares with the 5.3% increase for 2021 and the 143.5% gain for the same period last year.

[1] Real estate investment in China rose 3.7% in the January-February period, slowing from the 4.4% increase for 2021 and the 38.3% increase in the same two-month period last year. **New construction starts measured by floor area declined... , down 12.2%** from a year earlier, widening from the 11.4% fall for 2021.

Pipeline output: 40 accepted, 0 rejected — 5 strong, 35 weak

*Unlike the first two examples, one article here fans out across a large slice of the universe. This is the pipeline's **breadth behavior** — we need to check each tag to decide whether the reach is warranted.* The next slide bins all 40 tags by the quality of the transmission channel.

Example 4b — the 40 tags, binned by transmission quality

Tier 1 — Direct <i>Strong</i>	LME Copper (0.80), Hang Seng Index (0.75), Copper HG (0.75), LME Zinc (0.75), AUD (0.75) China is the world's largest consumer of industrial metals; construction is the primary end-use (wiring, plumbing, cladding, galvanized steel). HSI is dominated by Chinese property developers and their banks. AUD captures the commodity-FX channel — China drives iron-ore and coal demand, Australia's largest exports.
Tier 2 — Regional & base-metal complex <i>Defensible</i>	LME Aluminium, LME Nickel, LME Lead, Materials Sector, Kospi 200, SPI 200 Australia, Mini Nikkei, Mini TOPIX, CAD, 10Y/3Y Aus. Govt. Bonds Korea/Japan export capital goods, chemicals and steel to China; Australia's terms of trade hinge on iron ore and coal demand from Chinese construction. Real macro linkages; weak/low scores are appropriate because the channel is indirect.
Tier 3 — Global growth & risk sentiment <i>Stretched</i>	Brent, WTI, Gold, DAX, Mini S&P 500, MSCI Singapore, OMXS30 Sweden, NOK, 10Y JGB, 2Y/5Y/10Y/30Y UST & Ultra T-Bond The pipeline leans on "China is a global growth driver" and routes the shock into generic risk-off / commodity-demand channels. Coherent but exactly what the prompt warns against as "generic macro atmosphere." Correctly scored ≤ 0.45 .
Tier 4 — Tenuous <i>Questionable</i>	Soybeans, Cotton, Silver, JPY, 90-Day Aus. Bank Bill Multi-step chains (property → consumer confidence → meat → soy-meal). These survive validation because the prompt was deliberately loosened to favor over-tagging — a downstream relevance-score + source filter is the intended place to handle these.

Example 4c — zoom + a filtering recommendation

LME Copper — strong, 0.80, both (Tier 1)

AM: “China is the world’s largest consumer of copper, and the construction/real estate sector is a primary driver of industrial copper demand (wiring, plumbing, structural components)... transmits directly to copper prices via a reduction in projected industrial demand.” **Validator:** “... a direct bearish trigger for LME Copper demand.”

Soybeans — weak, 0.45, paragraph_only (Tier 4)

Validator: “... a sustained slump in property acts as a drag on the broader economy, which can lead to reduced consumer confidence and lower demand for protein (meat), particularly pork. Since pork consumption drives soybean-meal demand, and China is the largest soybean importer, this is a real but indirect channel.”

A five-step chain (property → confidence → meat → pork → soy meal) — but note: **China is genuinely the world’s largest soybean importer**, and the pipeline is not hallucinating facts. Whether this kind of macro-chain tag is *useful* for a PM is a judgment call a macro trader would have to make.

Recommendation: post-hoc filter

Drop any mapping that is **weak and score ≤ 0.45 and single-source** (i.e. only one of the two mappers fired). Intuition: a tag without cross-mapper corroboration *and* a sub-median score is exactly the “coherent but generic” bucket. On this article, the filter shrinks the tag set from **40** → **16**: it keeps all 5 Tier-1 strongs, the defensible regional/base-metal cluster (Materials, Kospi, LME Nickel/Lead, SPI 200), and the Tier-3/4 tags where both mappers independently agreed. It drops the lone-mapper “global growth” chains (gold, US Treasuries, EAFE, Sweden, JPY, Singapore, ...).

Bottom line: reasoning is grounded, not hallucinated; open question is calibration. The filter above is worth testing before tightening the validator prompt itself.

Example 5a — the “tags everything” article

Headline: *Dollar Firms, Ukraine War Should Send It Higher — Market Talk*

(DJNW, 2022-03-07, shard a)

Full article text

[0] 1344 ET — The dollar strengthens 0.4% against the euro and yen and the WSJ Dollar Index gains 0.6%. The Ukraine war should lead to a stronger dollar via at least three channels, Piper Sandler says: **safe-haven flows into US assets, a favorable US interest rate differential and a relatively better US economic outlook**. “Normally, the dollar peaks soon after a decisive Fed policy response.”

[1] “This crisis, however, is different because the **Fed is in a tightening mode given high inflation**.” A stronger dollar tightens US financial conditions, the firm says, but also makes imports cheaper for the US, which could help counter strong inflation pressures. “In the end, we think a stronger dollar will be another factor that could restrain the Fed this year.”

Pipeline output: 93 accepted, 0 rejected — 2 strong / 91 weak — *fans across every asset class*

Class distribution: commodity 25, equity index 23, gov’t bond 18, US equity sector 10, currency 9, STIR 6, volatility 2.

The 2 strong tags are two of the three assets the article names: Euro (0.95) and JPY (0.85). 3M SOFR is weak — arguably correct since the article discusses Fed *context*, not a direct SOFR trigger. The remaining 91 tags are weak “global macro atmosphere” inferences.

Example 5b — stress-test of the 4c filter

Applying the filter from Example 4c ($\text{weak} \wedge \text{score} \leq 0.45 \wedge \text{single-source}$) to this article:

Class	Before	After filter	Dropped
commodity	25	23	2
equity index	23	12	11
government bond	18	18	0
US equity sector	10	8	2
currency	9	9	0
short-term interest rate	6	5	1
volatility	2	2	0
TOTAL	93	77	16

What survives the filter is still too many (77 tags on a two-paragraph Market Talk blurb). The filter is deliberately gentle — the relevance-framed prompt increased cross-mapper agreement, so more tags now carry the `source=both` label and bypass the filter. Gov't bond, currency, volatility classes lose nothing because every tag in those classes is already both-source. The filter is *purely post-hoc* — no prompt re-engineering, no re-run needed.

Open question: should a two-paragraph blurb be allowed to tag 50+ assets at all? A hard cap would force the pipeline to surface its top picks.

Example 6a — a real failure we found, and fixed

Headline: *Nymex Light Crude Oil Close — Mar 23* (DJNW, 2022-03-23, shard c)

~10k chars of crude-oil contract data. No prose trigger anywhere in the body. The words “crude,” “oil,” and “cocoa” each appear **zero times**. Source: CME Group... Data as of: 15:54 ET

Contract Open High Low Last Chg Prev Settle
May '22 108.85 115.40 108.38 114.58 5.31 109.27
Jun '22 105.84... Dec '27 68.20...

BEFORE the fix (10 tags)

Strong @ score = 1.0:

Cocoa, Coffee, Kansas Wheat, Live Cattle, Sugar, Chicago
Wheat

Weak: Cotton, Feeder Cattle, Lean Hogs, Heating Oil

Crude Oil: 0 tags

All six strong tags were hallucinated. The ParagraphMapper's reasoning for Cocoa (verbatim): *“a direct data feed of settlement prices and trading activity for Cocoa futures contracts on the CME/NYBOT.”* Validator accepted every mapping.

Companion recovery: *ICCO Cocoa Daily Price* (2022-03-21, shard c) went 0 → **Cocoa strong 0.95**; *ICE Europe White Sugar Close* (2022-03-21, shard c) 0 → 1 tag. Details next slide.

AFTER the fix (2 tags)

WTI Crude Oil, strong, 1.00, paragraph_only

Brent Crude Oil, weak, 0.55

Hallucinated commodities: 0

Every one of the six wrong strong tags is gone. The pipeline now correctly tags the article's own asset (**WTI Crude Oil**) as strong, alongside a weak Brent tag for the adjacent benchmark. Perfect routing.

Example 6b — root cause and the two-part fix

What the validator was actually seeing

Pipeline audit: **no stage saw the headline.** `run_summarize`, `run_article_level`, `run_paragraph_level`, and the validator's `.build_validation_input` all constructed the article body as paragraphs-only. The headline field was read from the article dict only when writing the output JSON. So when the validator was asked *"is this article relevant to Cocoa?"*, it saw: a numerical price table, the asset label Cocoa, and a confident ParagraphMapper prior. Nothing contradicted the story, so it accepted. Six times.

Fix part 1: prepend [HEADLINE] to every stage

Prepend `[HEADLINE]` `<title>` to the input of every stage, plus one neutral prompt sentence. No rejection instruction — existing rules cover it. Once grounded on the headline, the ParagraphMapper can no longer claim a number table is Cocoa settlement data when the title says *"Nymex Light Crude Oil Close."*

Fix part 2: drop the "past price reporting alone" exclusion

The prompt's *WHAT IS NOT A SIGNAL* list had a bullet rejecting "past price reporting alone," meant to block secondary news mentions like "gold rose 2% yesterday" — but the mappers were also applying it to *primary* exchange settlement data, so ICE Sugar / ICE Gasoil / ICCO Cocoa got zero tags on their own-asset articles. Removed the bullet; the pipeline now treats an exchange's own close report as a valid relevance signal for the named asset (Bryan: relevance, not alpha).

Combined effect: strong tags 525 → 727, both-source tags 1987 → 2428, and 34 of 57 previously zero-tag non-company articles now fire at least one tag.

Company-specific gate

Does the Stage 0 gate actually work?

725 filtered as company-specific (73%), 275 kept. The gate has to do two things: (a) drop pure corporate noise, and (b) *not* drop articles that feature a company but contain real macro content.

Correctly gated out (5 of 725)

- *CEO Herrman Acquires 101,682 Of TJX Companies Inc* — insider buy
- *BSE: Ajanta Pharma Ltd. — Compliances Reg 39(3) Details Of Loss Of Certificate* — administrative filing
- *eMagin 4Q Loss/Shr 3c* — microcap earnings miss, no macro mention
- *Artiva Biotherapeutics Appoints Laura Bessen, M.D., to Its Board* — board appointment
- *HK Bourse: Announcement From SINOPEC Shanghai Petrochm* — regulatory disclosure, no substance

Also correctly gated: Moody's municipal-debt rating changes, capital raises (e.g. *Phoenix Copper Seeks to Raise \$60 Mln*), single-company Russia exits (*Rockwell Automation Suspends Business in Russia*).

Correctly kept (company in headline, macro inside)

- *Carnival 1Q Loss/Shr \$1.66* — earnings note that discusses Ukraine's impact on fuel prices → 10 tags (Brent, WTI, Gas Oil, Energy Sector, Cons. Discretionary, FX).
- *S&PGR: Rising Costs Loom As Key Test For Corporate Australia* — sector note citing RBA normalisation + cost inflation → SPI 200 *strong*, 90-Day AU Bank Bill *strong*, AUD, AU gov bonds.
- *Boeing Co.: Suspended Parts and Maintenance Support for Russian Airlines* — EU sanctions content → Industrials Sector, DAX, Euro Stoxx 50, VSTOXX.
- *Small Business Growth Surpasses Pre-Pandemic Levels* — Equifax Canada credit report → CAD, 10Y Canada Gov Bond, Cons. Discretionary, Energy.
- *Wabtec and BNSF Railway Biofuels Pilot* — renewable-diesel pilot → Energy, Heating Oil, Corn, Soybeans (biofuels feedstock).

Carnival earnings — macro hidden inside a company headline

Headline: *Carnival 1Q Loss/Shr \$1.66 > CCL*

(DJNW, 2022-03-22, shard c)

A naive company-name filter would drop this instantly (Carnival Corp, ticker, loss-per-share). The pipeline keeps it because Stage 0 extracts the macro substance in the body:

Stage 0 macro_summary (excerpt)

Carnival reported a US GAAP net loss of \$1.9B for Q1 2022, citing Omicron. **The company noted that the invasion of Ukraine is creating uncertainty and exerting upward pressure on fuel prices, which is materially impacting its business and liquidity.** Fuel consumption forecast for 2022 is 2.2M metric tons.

Downstream tags (10 accepted, all weak): Brent 0.45, WTI 0.40, Gas Oil 0.40, Energy Sector 0.35, Cons. Discretionary 0.55, EUR/GBP/AUD.

Why this proves the gate is doing real work:

- Not a keyword filter. Every surface feature screams “drop” — ticker, company name, earnings number — yet the article is kept because the body contains substantive fuel-price commentary tied to Ukraine.
- Downstream routing respects the economic mechanism: heaviest tags are on oil and fuel, *not* on cruise operators. Consistent with the article’s macro angle, not its corporate framing.
- All tags weak. Pipeline correctly treats earnings as a secondary source of fuel-price commentary, not a primary trigger.

Gate stress test — macro keywords in gated headlines

We sampled every gated article (cs=True) whose headline contained a macro keyword (Fed, CPI, Ukraine, Russia, oil, gold, copper, ...) — **26 articles** — and read the bodies.

Headline (gated)	Verdict
<i>Barrick Gold Corp. Stock Rises Tuesday</i>	Correct. MarketWatch auto-blurb of a single stock's price move. Matches the prompt's explicit non-signal: " <i>Past price reporting alone.</i> " 10 Barrick Gold blurbs in the sample, all gated.
<i>Rockwell Automation Suspends Business in Russia</i>	Correct. Body verbatim: " <i>Russia and Belarus account for less than 0.5% of total revenue.</i> " Article itself declares it immaterial. Contrast Boeing/titanium (not gated, 25% of world supply).
<i>ITAB Shop Concept discontinues Russia ops</i>	Correct. Retail-fixtue company closing a 125-employee facility. Not a bellwether.
<i>Phoenix Copper Seeks \$60 Mln for Empire Open Pit</i>	Correct. Single-company loan-note raise; \$60M for one mine is not a macro copper supply story.
<i>Manz AG: Strong demand for battery production</i>	Borderline. Press release about winning a Li-ion battery order. A macro strategist could argue battery demand is a real Cu/Ni/Li signal; the gate treats it as a single-company order. Defensible but the closest call in the 26.

Bottom line: 25 of 26 are clearly correct — all match the prompt's explicit "not a signal" categories. Manz AG is a genuine gray area, defensibly gated under the current rules. The gate reads *body content*, not headline keywords.

External benchmark — DJNW's own codes field

Every DJNW article ships with a codes dict. The company field lists tickers — ticker-based ground truth for “DJNW thinks this article is about a specific company.”

DJNW's own tagging: 67.3% of the 1000 articles have a company ticker. **Our gate:** 72.6% cs=True. **Within 5.3 pp.**

	Our gate = True	Our gate = False
DJNW has company ticker	640	33
DJNW no company ticker	86	241

Raw agreement 88.1%, recall 95.1% on DJNW-company articles. The 12% disagreement decomposes cleanly:

33 we keep despite a company ticker — the intended macro-in-body cases: Apple / Boeing / Sylvamo / Wintershall Russia sanctions; Adidas, Leumi, Arcos Dorados, Sembcorp, Taiheiyo Cement earnings with macro content; *Heard on the Street* columns; Wabtec biofuels; JD Logistics oil-price pressure; nickel surge + Chinese manufacturers.

86 we gate despite no ticker — DJNW's ticker field can't express these: 23 closing-price bulletins, 10 Moody's/Fitch/S&PGR municipal bond ratings, 10 exchange data reports, 8 small/private-company press releases, 3 WSJ culture pieces, 32 other. All defensibly non-macro.

Takeaway: 5pp from DJNW's own ticker rate; disagreement is entirely (a) deliberate macro-body keeps and (b) non-ticker corporate content.

Aggregate diagnostics

Relevance score distribution (all 4,019 accepted tags)

score	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95–1.0
weak (3292)	130	368	716	1108	318	494	55	68	29	3	.	.	.	.
strong (727)	.	.	.	.	.	.	.	.	6	120	83	164	78	276

Honest observations:

- **Anchoring at specific values.** Weak scores cluster at 0.40/0.45/0.55; strong at 0.75/0.85/0.95. Earlier experiments with guard text in the prompt ("do not default to 0.85") made this *worse*, not better — the mapper just anchored elsewhere.
- **The strong/weak boundary is clean.** Strong scores are bounded below at 0.70, weak bounded above at 0.75, with only minor overlap. The *label* is doing more work than the raw score.
- **Implication:** treat (signal, score) as a pair, not as a continuous variable. Downstream: bin into 3–4 buckets (strong/high, strong/low, weak/med, weak/low) rather than use scores as a float.

Validator catches a factual hallucination — 3-Month SARON

Article: *Global Forex and Fixed Income Roundup: Market Talk* (DJNW, 2022-03-02, shard a) — discusses BoC rate hike and Desjardins' 2022 BoC outlook. ParagraphMapper proposed **3-Month SARON** via a Canadian central-bank chain.

ParagraphMapper [paragraph 7] — the error

"The transmission channel is direct: **the Bank of Canada (BOC) controls the short end of the Canadian yield curve...** primary drivers for the 3-Month SARON (Swiss Average Rate Overnight, though in this context, *the user is likely referring to the Canadian equivalent or the specific short-term rate benchmark for Canada given the BOC context*)..."

Validator — **REJECTED**

"The asset is '3-Month SARON', the Swiss Average Rate Overnight — **the benchmark interest rate for Switzerland**. The ParagraphMapper's claim in paragraph [7] that **the BOC controls the '3-Month SARON' is factually incorrect; the BOC controls the Canadian overnight rate (CORRA), not SARON**. There is no mention of the Swiss National Bank, Swiss inflation, or Swiss economic conditions in the article."

Why this matters: the ParagraphMapper even flags its own uncertainty ("the user is likely referring to the Canadian equivalent") but still proposes the mapping. The Validator — the *same* Gemma 4 26B A4B model, re-prompted — catches the factual error, names the correct central bank (SNB) and the correct Canadian rate (CORRA), and rejects. Multi-stage self-correction as intended.

Source × accept/reject — what the validator actually does

Each (article, asset) pair reaches the validator with a **source** label recording which mappers fired:

Source	Accepted	Rejected	Total	Reject rate
both (AM $\cap$ PM agreed)	2,428	12	2,440	0.5%
article_only (only AM fired)	850	33	883	3.7%
paragraph_only (only PM fired)	741	93	834	11.2%
total	4,019	138	4,157	3.3%

Read-outs:

- **Cross-mapper agreement is an extremely strong prior.** When both mappers independently flag the same asset, the validator rejects just 0.5% of the time — the two mappers effectively pre-corroborate each other.
- **The ParagraphMapper is still the noisiest component.** Its lone tags get rejected 22× more often than two-mapper tags, which is the empirical basis for the “single-source filter” recommendation from Example 4c.
- **The ArticleMapper is better calibrated solo** (3.7% reject), which makes sense: it sees the whole article, not one paragraph stripped of context.

Asset universe coverage

All 95 assets are tagged at least once, but the distribution is long-tailed.

Class	# assets	Tags	Strong	Tags/asset	Strong/asset
commodity	25	1,127	300	45.1	12.0
equity index	23	882	133	38.3	5.8
government bond	18	882	148	49.0	8.2
currency	9	549	44	61.0	4.9
US equity sector	11	350	49	31.8	4.5
short-term interest rate	7	121	29	17.3	4.1
volatility	2	108	24	54.0	12.0

Head (most-tagged): CAD (103), Brent (87), WTI (81), Gold (81), AUD (78), Euro (77), Heating Oil (77), Energy Sector (70), Euro Bobl (69), ICE Gas Oil (68).

Tail (4–11 tags over 1000 articles): 3-Month SARON (4), Health Care Sector (5), Cocoa (5), 90-Day NZ Bank Bill (5), Coffee (7), Cotton (10), Comm. Services Sector (11).

Observations: energy complex and G10 FX dominate (predictable for March 2022); the tail is regional short rates and niche softs; STIR has lowest tags/asset because SOFR/SONIA/ESTR dominate; volatility 54 tags/asset is just VIX + VSTOXX.

Cost & scale

Per-stage call budget (baseline measurement)

Pipeline calls for 1000 DJNW articles:

Stage	What it does	LLM calls	Wall-clock
Stage 0 — Summarize	1 call per article (macro summary, company gate)	1,000	1:56
Stage 1 — ArticleMapper	1 call per (non-company article $\times$ 95 assets)	26,125	3:53
Stage 2 — ParagraphMapper	1 call per (paragraph $\times$ 95 assets)	107,065	10:26
Stage 3 — Validator	1 call per candidate (article, asset) pair	3,484	2:18
TOTAL		$\sim$137,700	18:57

Throughput: $\sim$ 121 calls/sec on $1 \times$ B200, batched through vLLM 0.19 with prefix caching. **Stage 2 is 78% of all calls** — the dominant cost and the obvious lever for scaling. The next slide translates this into wall-clock estimates for the full corpus.

Full-DJNW scaling — 1996+, Bouchet fleet

Corpus: 29.11M DJNW articles (1996-01 → 2025-08); trimming the pre-1996 head (2.56M of mostly headline-ticker feeds) removes ~8% of the total workload. Full corpus (1979–2025) = 31.67M for reference.

Baseline: 1.137 sec/article end-to-end on 1× B200, with the 73% company-specific gate already baked in. 1996+ workload is ~**9,200 GPU-hours**; full corpus is ~10,000.

Bouchet fleet: gpu_b200 (56 B200s, ~3,170 art/hr/GPU measured), gpu_h200 (72 H200s, ~2,800 est.), gpu_rtx6000 (56 RTX Pro 6000 Blackwells, ~2,400 measured). **Fleet total ~184 GPUs**, pipeline embarrassingly parallel across month shards.

Scenario (1996+ corpus)	Wall clock	w/ Stage-2 opt.
1× B200 baseline	383 days	191 days
8 B200s (small sbatch)	48 days	24 days
16 B200s	24 days	12 days
32 B200s (fair share)	12 days	6.0 days
Full B200 partition (56)	6.8 days	3.4 days
All H200s (72)	6.0 days	3.0 days
Full fleet (184 mixed)	2.4 days	1.2 days

Stage 2 is the dominant cost lever. Two realistic levers: asset-universe trim (~15–20%, pending Bryan) and a paragraph-level prefilter. The “smaller Stage-2 model” idea did *not* pan out — E4B ablation next.

Small-model ablation program

Question: can a smaller model replace Gemma 4 26B A4B on Stage 2 without tanking quality? Same 1000 DJNW articles, seed 42, same prompts.

Model	Architecture	Why tested	Result
Gemma 4 26B A4B (<i>baseline</i>)	MoE, 4B active, server-tuned	<i>baseline</i>	18:57 wall clock, 4019 acc, 727 strong, 138 rej
Gemma 4 E4B	dense 4B, edge-NPU (PLE, selective activation)	drop-in speedup attempt	failed: 33:33 wall, +84% tags, 5× more strong, cross-currency hallucination (SARON on BoE)
Qwen 3.5 9B	dense 9B, server-tuned	non-Gemma quality datapoint; not a speed candidate	[pending — running on B200, gate verified 72.9%]
Qwen 3.5 4B	dense 4B, server-tuned	true speedup candidate; isolates architecture vs param count	[pending — queued on RTX]

Why E4B failed: 26B A4B is already at the 4B-active compute floor, so “4B edge dense” brings no speedup. Edge-NPU architectures are tuned for mobile efficiency, not server tensor cores. Quality also collapsed — pipeline is reasoning-bound.

What Qwen 3.5 will tell us: if 4B server-dense works → E4B failed due to edge tuning, not param count. If it fails → reasoning floor is above 4B regardless of architecture, and 26B A4B is irreducible for this task.

DJNW input shapes — a running taxonomy

Shape	Example	Pipeline behavior
Narrative analysis column	<i>Heard on the Street: Russia Can't Fly Without The West</i>	Multi-paragraph WSJ commentary. Best-case input: evidence localization + domain routing (titanium → Materials).
Market Talk blurb	<i>BOE's Rate Decision Could Benefit Sterling (HSBC)</i>	Single-paragraph analyst quote, ~500–1000 chars. Handled well; AM and PM converge.
Market news update	<i>China Jan-Feb Property Investment Rose 3.7%</i>	Short news story with data + commentary. The “breadth” case — fans across 40+ tags; post-hoc filter handles the tail.
Agricultural data release	<i>USDA U.S. Export Sales and Shipments: Soybeans</i>	Structured country-by-country export table. Evidence localized to the data paragraph (0.95) vs the header (0.50).
Exchange settlement report	<i>ICE White Sugar Close, Nymex Crude Oil Close, Comex Gold Close</i>	Pure numeric tables, no prose trigger. Was the canonical failure case (hallucination or silence); post-fix correctly tags the named asset (e.g. WTI 1.00 strong on the Nymex article).
Benchmark yield snapshot	<i>10-Yr Benchmark Govt Yields — U.S. vs Other Nations</i>	Cross-country yield table. Weak-signal data point across relevant sovereigns + FX.
Market roundup / digest	<i>Global Forex and Fixed Income Roundup: Market Talk</i>	Multi-topic, multi-region aggregation. Dangerous cross-region reasoning — where the SARON factual catch lives . Validator is the backstop.
Company earnings w/ macro	<i>Carnival 1Q Loss/Shr \$1.66</i>	Corporate headline, macro body (Ukraine fuel pressure). Gate correctly keeps it; routing goes to oil/fuel complex, not just Cons. Discretionary.
Pure corporate event	<i>CEO Herrman Acquires 101k Of TJX; HK Bourse Announcement; BSE Compliance</i>	Insider trades, regulatory filings, MarketWatch auto-blurbs. Stage-0 gate drops 73% of the DJNW universe in this bucket.

This list is not exhaustive — it's a running inventory of shapes we have verified so far. New shapes will surface as we expand to other months, other wire services, and eventually other text corpora (earnings transcripts, central bank speeches, research notes).